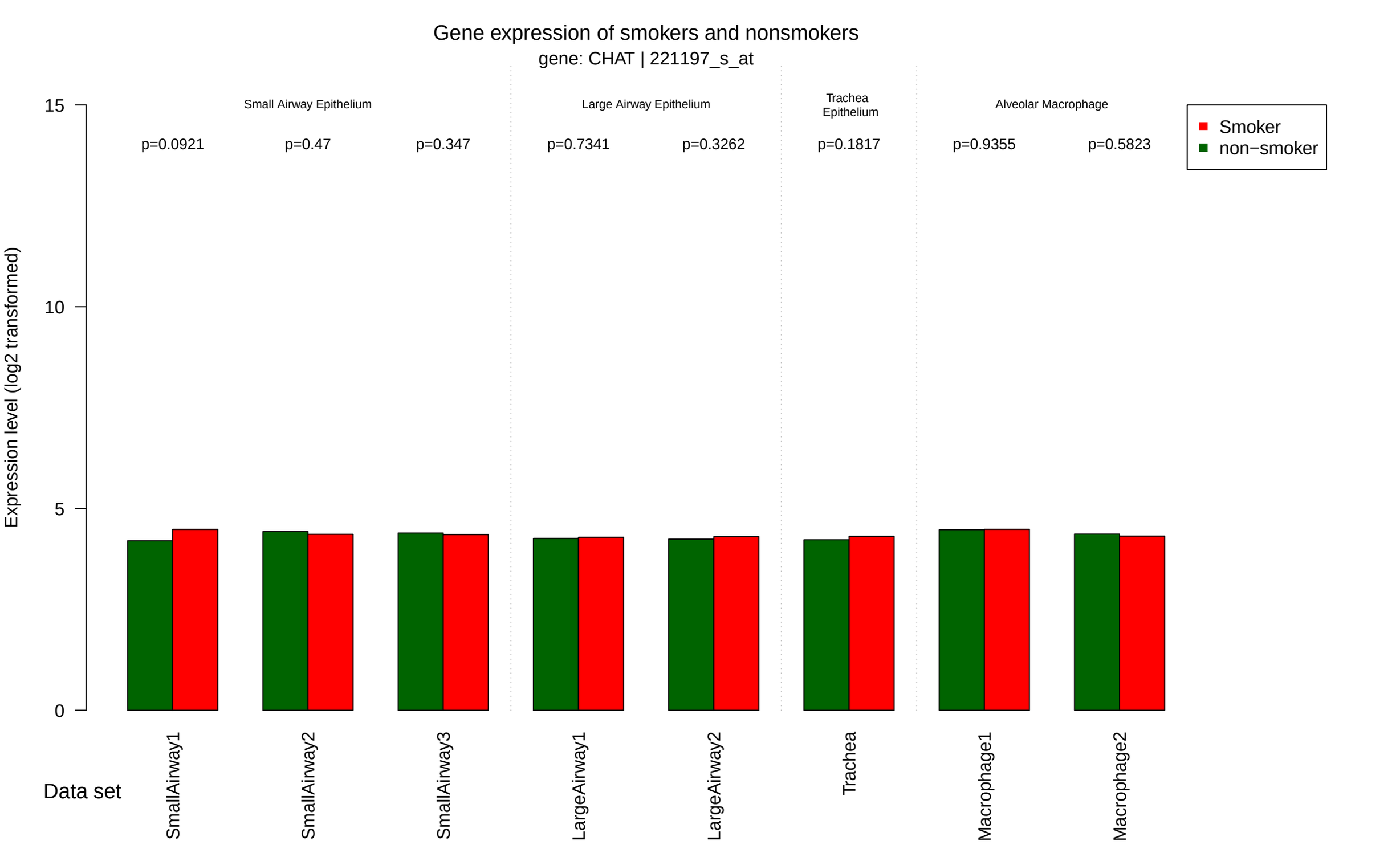
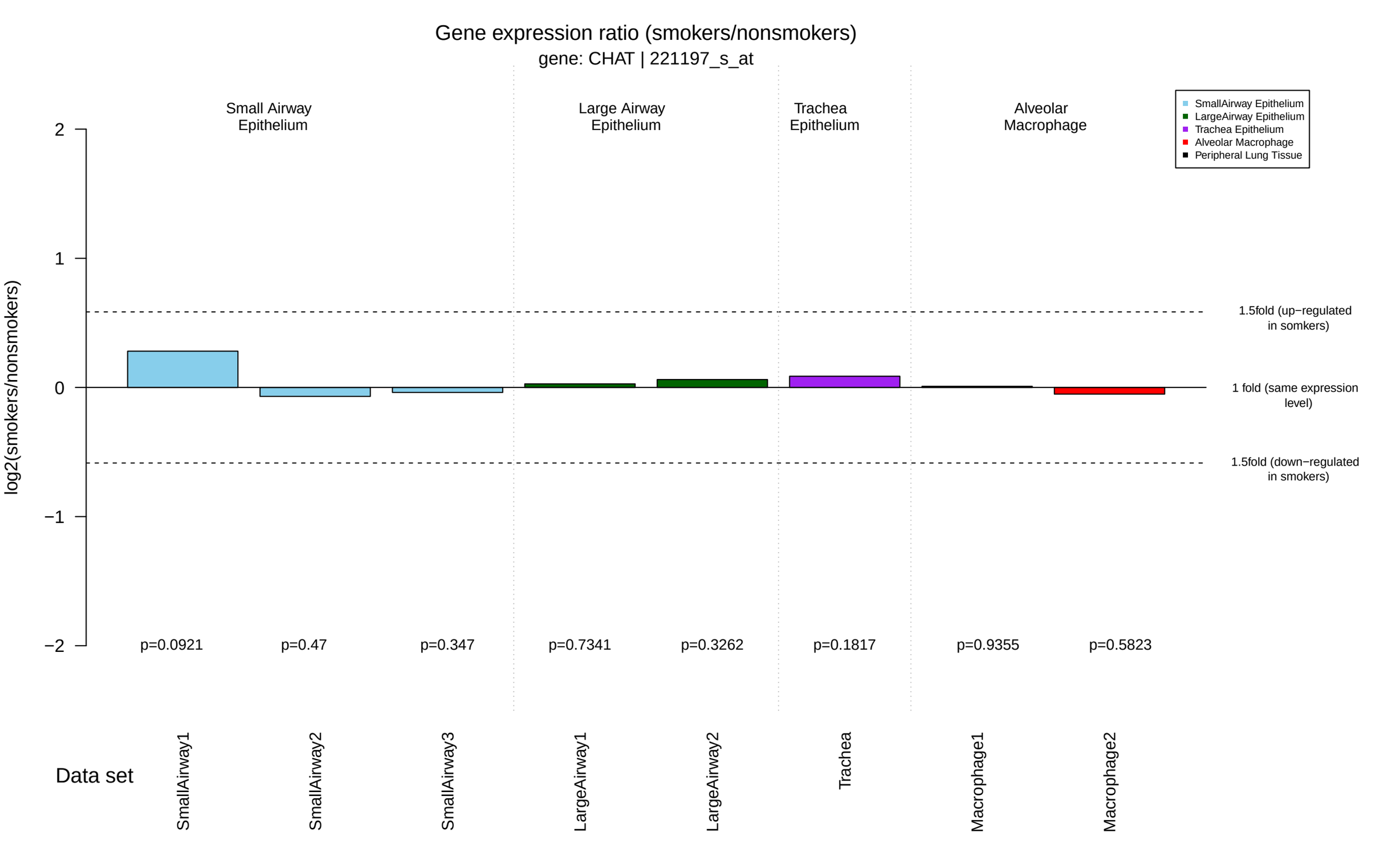
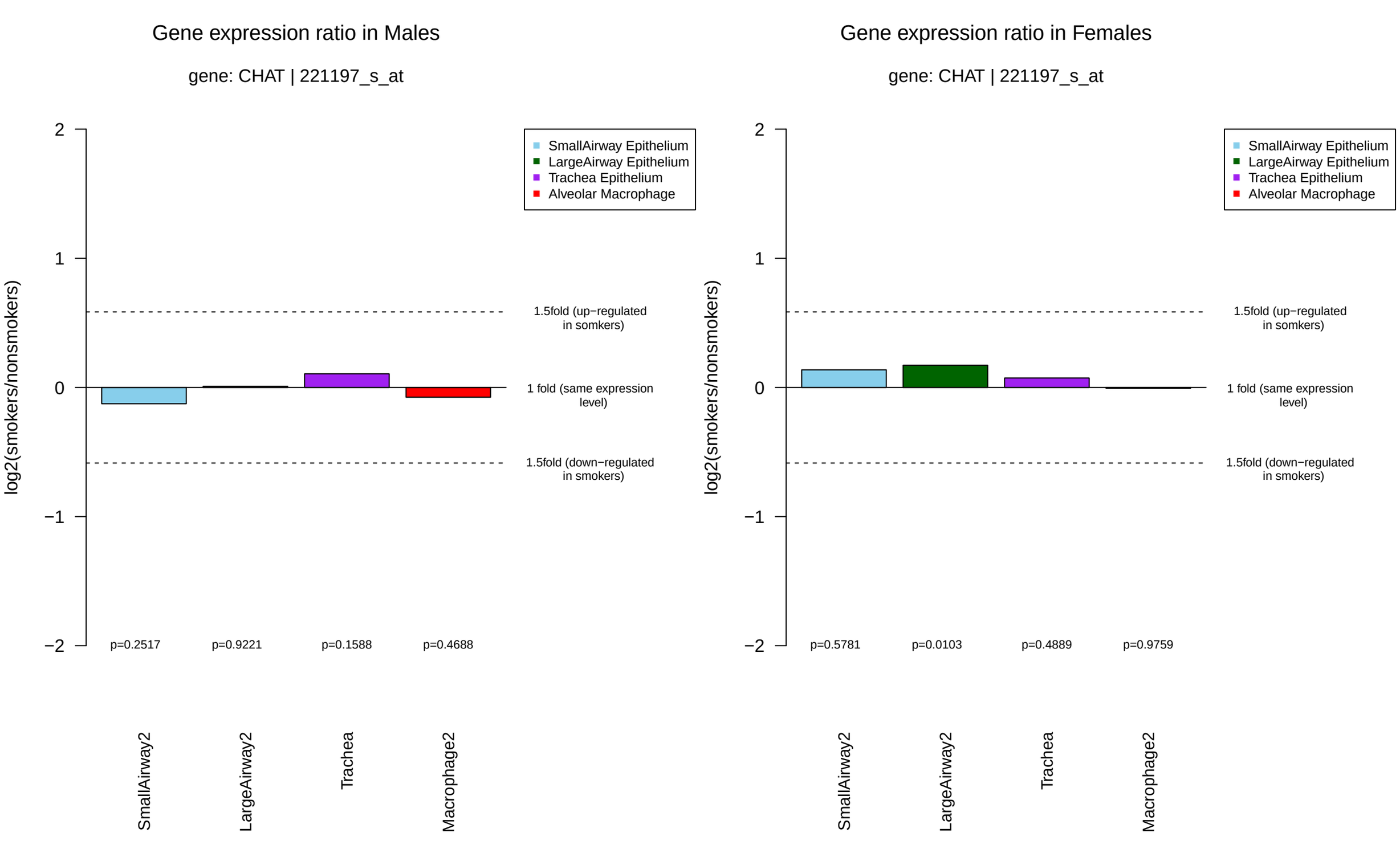
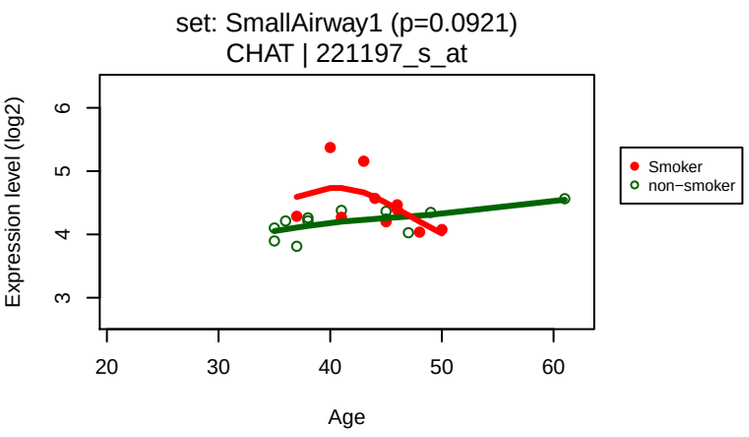
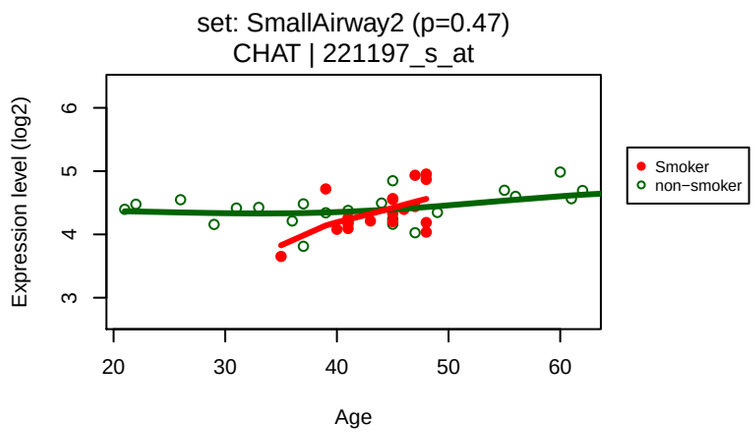
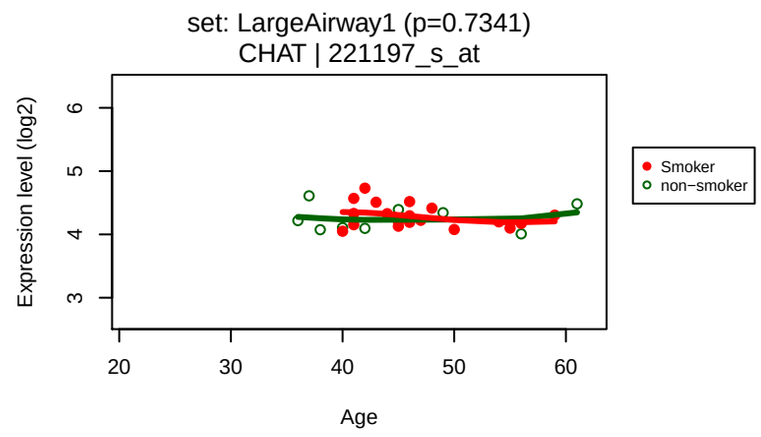

Gene expression of smokers and nonsmokers
gene: CHAT | 221197_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0921
p=0.47
p=0.347
p=0.7341
p=0.3262
p=0.1817
p=0.9355
p=0.5823
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

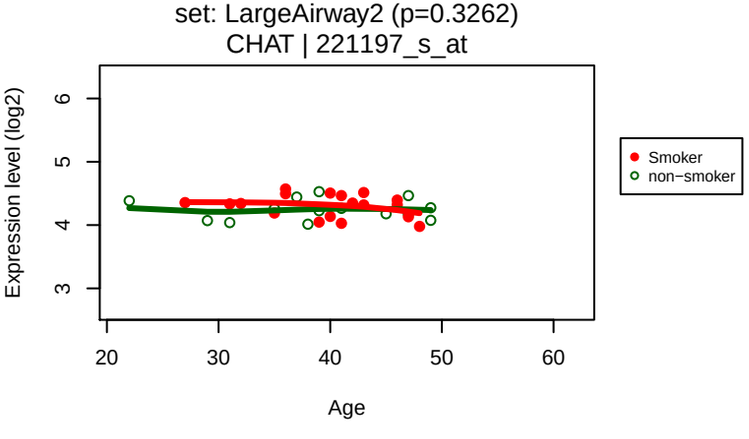

Gene expression ratio (smokers/nonsmokers)
gene: CHAT | 221197_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0921
p=0.47
p=0.347
p=0.7341
p=0.3262
p=0.1817
p=0.9355
p=0.5823
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

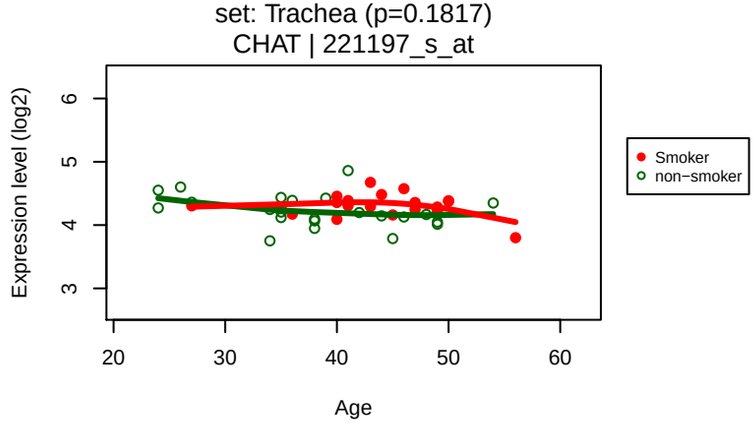

Gene expression ratio in Males
gene: CHAT | 221197_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2517
p=0.9221
p=0.1588
p=0.4688
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHAT | 221197_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5781
p=0.0103
p=0.4889
p=0.9759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

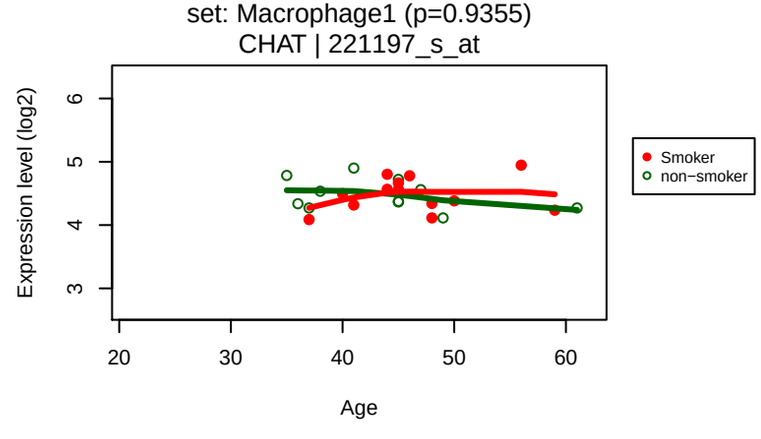

set: SmallAirway1 (p=0.0921)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker

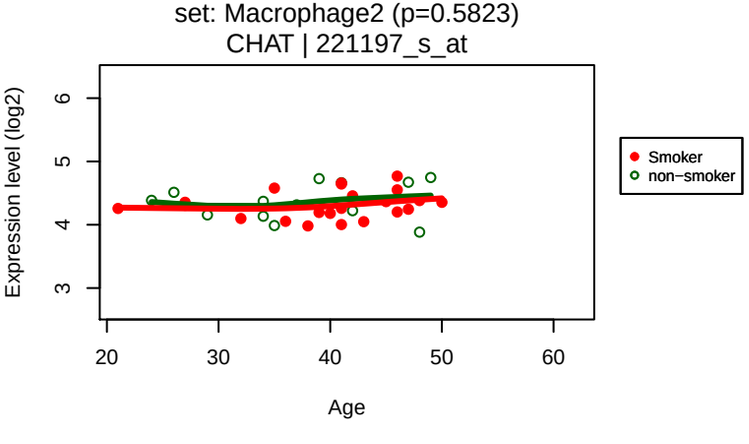

set: SmallAirway2 (p=0.47)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker

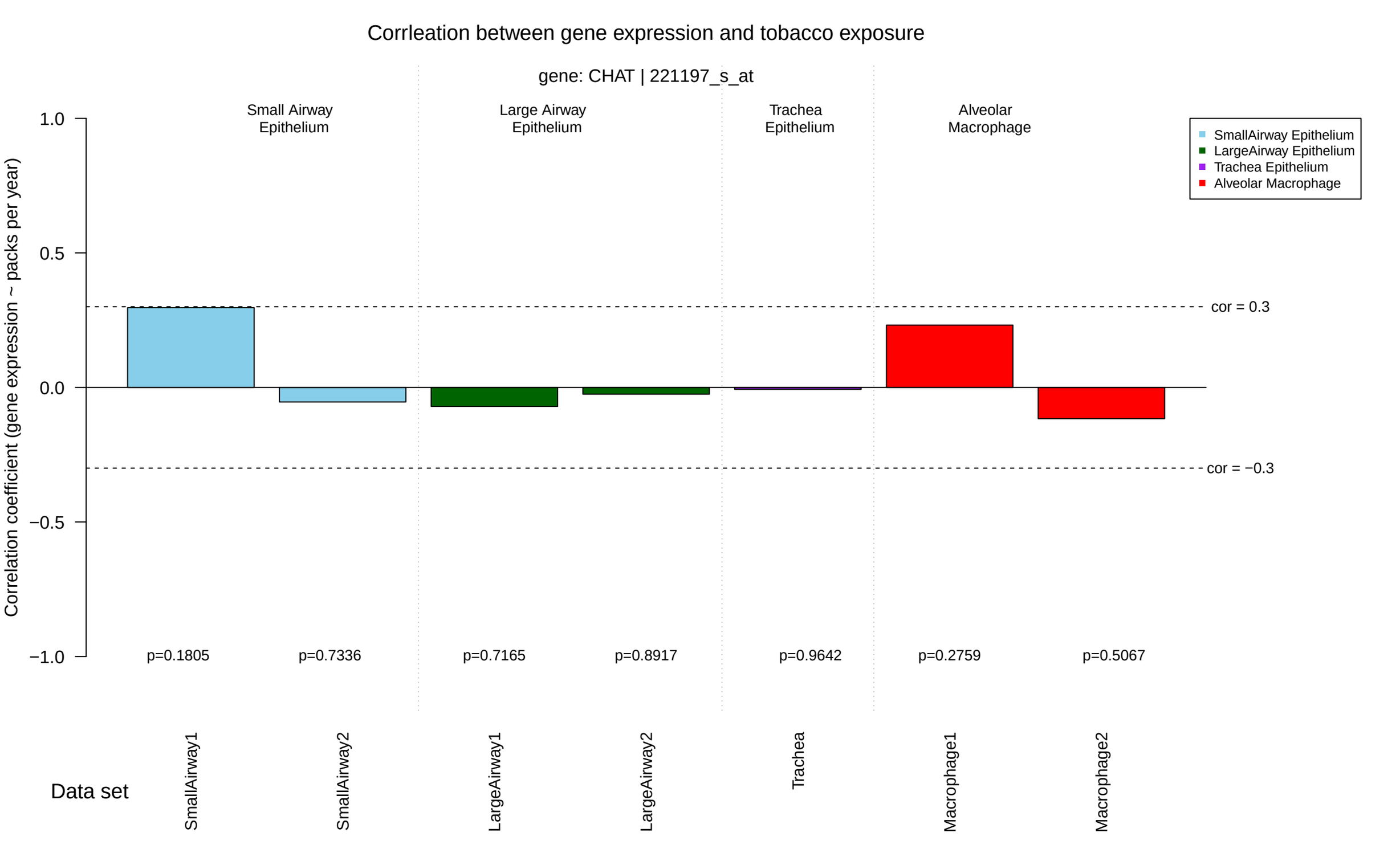

set: LargeAirway1 (p=0.7341)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3262)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1817)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9355)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5823)
CHAT | 221197_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHAT | 221197_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1805
p=0.7336
p=0.7165
p=0.8917
p=0.9642
p=0.2759
p=0.5067
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2